

Array (Search)

arr = [2, 8, 9, 6, 4, 1]
0 1 2 3 4 5

3rd Element $\Rightarrow$ Index arr[2]
TC $\rightarrow O(1)$

LinkedList (Search)

① $\rightarrow$ ② $\rightarrow$ ③ $\rightarrow$ ④ $\rightarrow$ null

3rd Element $\Rightarrow$ Traverse
n elements $O(n)$ slow

Array (Insert)

arr =

1	2	4	5	6
0	1	2	3	4

1	2	3	4	5	6
0	1	2	3	4	5

shift $\rightarrow$

$O(n)$ (slow) $\rightarrow$

LinkedList (Insert)

① $\rightarrow$ ② $\rightarrow$ ③ $\rightarrow$ ④ $\rightarrow$ null
② $\downarrow$ ① $\uparrow$ ③

3 steps (finite)

$O(1)$ constant

(Changes)

① $\rightarrow$ ② $\rightarrow$ ③ $\rightarrow$ null

T1 1 2 1

T2

1 3 4
4

(* reverse) (Stack) (LIFO)

4
3
1

 $\rightarrow$ top
empty

true

4
3
1

 $\rightarrow$ top

false

$\rightarrow$ s1 $\rightarrow$ $\{ [()] \}$

s2 $\rightarrow$ $[\{] (\{)$

Balance Parentheses $\rightarrow$

pseudo code:

if $S(i) == \{ or (or [$
push to stack;

else $S(i) == \} or) or]$
or $\rightarrow]$ s.top [$\rightarrow)$ s.top ($\rightarrow]$ s.top { false
s.pop

4	
3	{
1	[

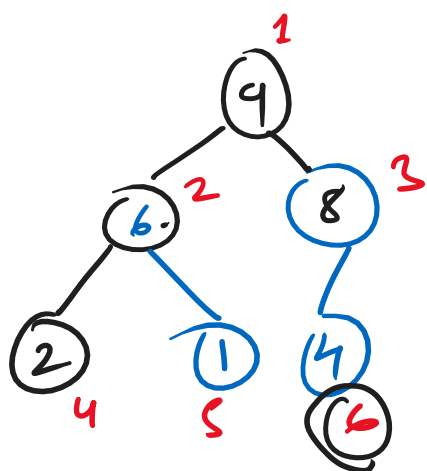
$] s.top \{$ false

empty $\rightarrow$ true

$\{ [\{]$
 $\{ \}$

(log N) int arr = [8, 2, 9, 6, 1, 4]
0 1 2 3 4 5

Max Heap: ($L < N > R$)



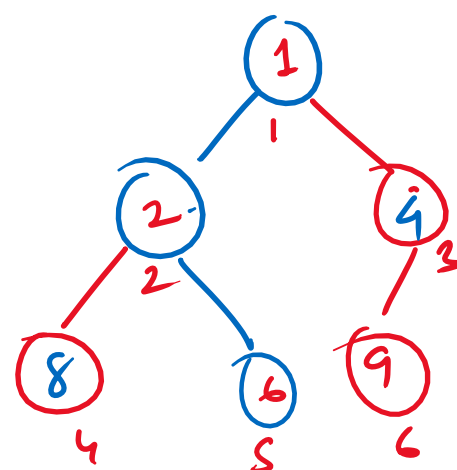
Max Heap

-1	9	6	8	2	1	4
0	1	2	3	4	5	6

$\downarrow$ Sentinel / cardinal value

Min Heap: ($L > N < R$)

$i \rightarrow lci \ \& \ rci$
 $lci = 2 * i$
 $rci = 2 * i + 1$
 $i \rightarrow$ parent
 $\downarrow$ or p_i
 $\frac{i}{2}$



Min Heap

-1	1	2	4	8	6	9
0	1	2	3	4	5	6

CBT
 $L \rightarrow R$
Convert level
can be empty
 $O(1)$